Quantification: 536-63x_Hyp_RD_z03c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2413.42	2332	1414.85	px^2
major_axis_length_px	lacuna shape	93.7767	89.3059	45.516	px
minor_axis_length_px	lacuna shape	36.2309	33.9956	10.5225	px
aspect_ratio	lacuna shape	2.484	2.5713	0.7633	unitless
eccentricity	lacuna shape	0.8772	0.9203	0.0999	unitless
solidity	lacuna shape	0.8024	0.8337	0.1144	unitless
roots_count	headline	10.25	8	8.3353	count
ring_length_r30_px	headline	311.583	254.5	239.647	px
ring_length_r60_px	reported	618.5	551.5	432.869	px
owned_length_px	ownership-dependent	1051.21	621.071	1223.99	px
edge_count	ownership-dependent	50.1667	30.5	60.6013	count
mean_edge_length_px	ownership-dependent	22.5532	21.8942	7.0956	px
perimeter_px	normalised	290.285	235.622	177.064	px
ring_area_r30_px2	normalised	9790.25	8516	3690.8	px^2
ring_area_r60_px2	normalised	24031.7	22415.5	6739.69	px^2
in_frame_fraction_r30	normalised	0.9777	1	0.0773	unitless
in_frame_fraction_r60	normalised	0.9689	1	0.1021	unitless
ring_density_r30	normalised	0.029649	0.0301761	0.0124233	px^-1
ring_density_r60	normalised	0.0242617	0.0249305	0.0106189	px^-1
roots_per_100px_perimeter	normalised	3.5319	3.5983	1.7904	per 100 px
ring_attached_length_r30_px	network v2	273.25	206.5	251.267	px
ring_attached_length_r60_px	network v2	460.75	352.5	426.916	px
ring_length_w_r30_px	network v2	353.978	286.016	275.795	px
ring_length_w_r60_px	network v2	700.605	633.293	496.364	px
sholl_crossings_r10	network v2	7.25	4.5	6.6212	count
sholl_crossings_r20	network v2	7.75	7.5	4.9198	count
sholl_crossings_r30	network v2	8.25	7.5	5.8016	count
loop_count	loops, report only	2.5833	0.5	4.7378	count
loop_edge_share	loops, report only	0.0929	0.0513	0.1198	share
ring_width_mean_r30_px	width	7.0099	7.2027	1.0085	px

Per field
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measure	value	units
analysed_area_px2	1.0029e+06	px^2
total_skeleton_length_px	21545	px
canalicular_length_density_per_px	0.0214828	px^-1
junction_count	392	count
junction_density_per_px2	0.00039087	px^-2
lacuna_count	12	count

lacunae_per_px2	1.197e-05	px^-2
skeleton_component_count	387	count
mean_component_length_px	55.6718	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0242897	px^-1
field_density_without_flagged_per_px	0.0216375	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	834	px
unattached_component_count	43	count
canalicular_length_density_attached_per_px	0.0214828	px^-1
field_length_density_with_unattached_per_px	0.0223144	px^-1
excluded_area_px2	16718	px^2
analysed_field_area_px2	1.03186e+06	px^2
loop_count	38	count
loop_edge_share	0.1188	share
edge_count	1162	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6216	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	21296	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	4424	168.508	61.7794	2.7276	0.9304	0.5066	-0.0985	178.037	485.058	False
2	740	35.5513	27.759	1.2807	0.6248	0.9091	-0.3246	161.099	832.037	False
3	3218	123.087	35.0225	3.5145	0.9587	0.8067	-0.0876	285.404	998.443	False
4	4546	152.562	46.2815	3.2964	0.9529	0.6884	-0.3441	401.046	256.605	False
5	3934	133.278	41.5402	3.2084	0.9502	0.7505	-0.3234	452.159	836.983	False
6	2906	117.901	38.3023	3.0782	0.9458	0.7758	-0.3279	473.994	376.288	False
7	2475	101.315	32.9688	3.0731	0.9456	0.8624	-0.3801	520.411	501.561	False
8	2189	73.1101	40.4728	1.8064	0.8328	0.9132	-0.5214	637.506	936.232	False
9	1176	57.7542	28.7935	2.0058	0.8669	0.8276	-0.2231	683.004	402.083	False
10	679	37.9692	24.2376	1.5665	0.7697	0.8573	-0.1349	734.916	174.795	False
11	829	46.9885	25.6067	1.835	0.8385	0.8399	0.1493	873.288	867.089	False
12	1845	77.2965	32.007	2.415	0.9102	0.8909	0.0147	929.902	369.506	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	32	973	1781	4336.05	215	20.1677	7.1588
2	7	229	496	1170.23	48	24.3799	8.4723
3	7	192	439	493.818	22	22.4463	6.537
4	18	483	961	2562.75	124	20.6673	7.1413
5	15	402	732	894.235	34	26.301	7.2466

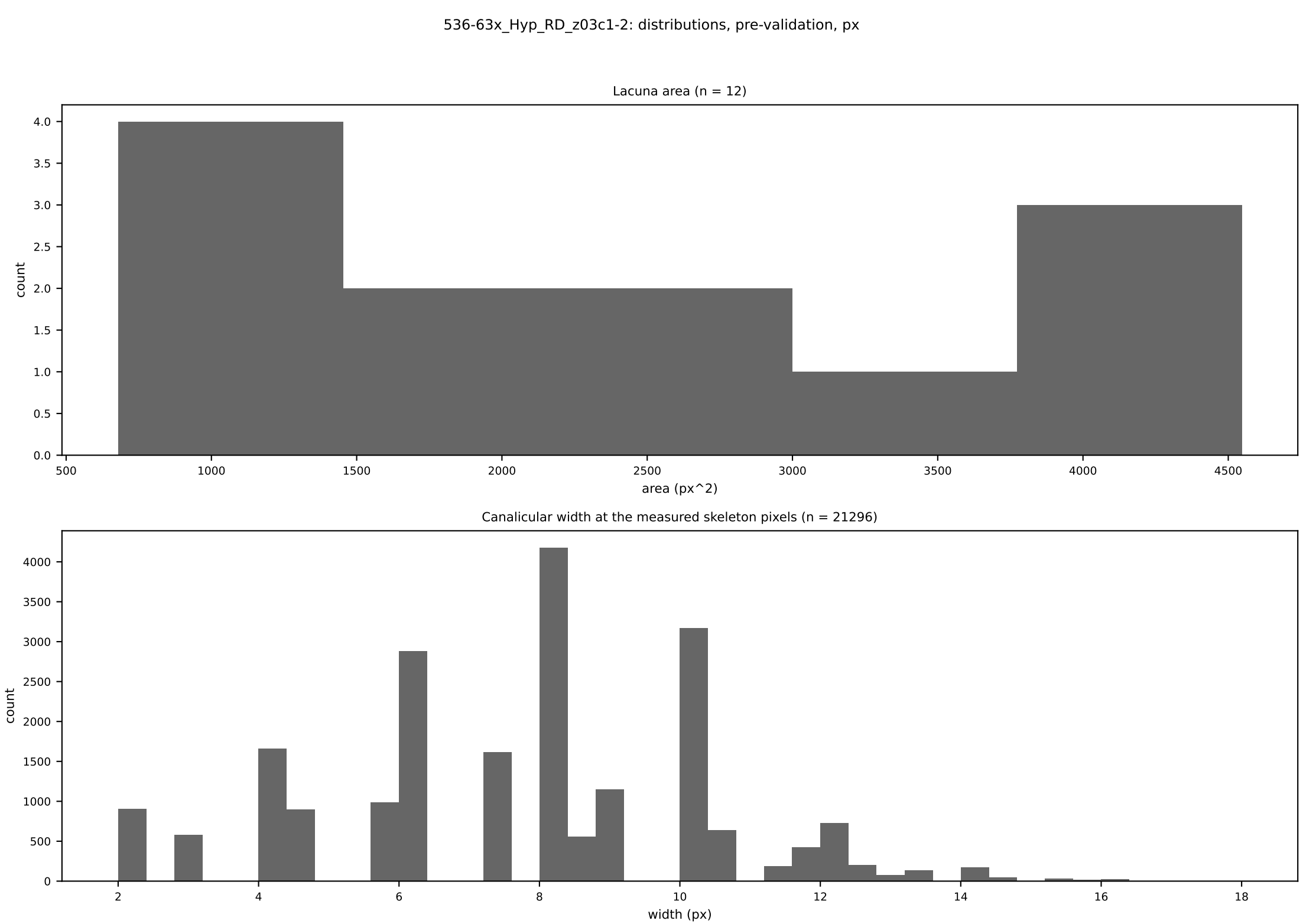
6	11	326	615	598.985	27	22.1846	8.1349
7	8	280	621	518.76	22	23.58	7.3812
8	2	116	171	63.8701	5	12.774	4.8013
9	8	288	579	643.156	42	15.3132	7.5747
10	8	217	524	842.553	39	21.6039	7.531
11	1	44	90	41.7279	1	41.7279	5.8562
12	6	189	413	448.333	23	19.4927	6.2838

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	733.068	18106	39240	1	1	0.0537391	0.0453874	4.3652
2	114.811	5937	17499	1	1	0.0385717	0.0283445	6.097
3	328.049	8327	18178	0.7322	0.645	0.0230575	0.0241501	2.1338
4	472.659	14091	31679	1	1	0.0342772	0.0303355	3.8082
5	380.534	12648	30304	1	1	0.0317837	0.0241552	3.9418
6	324.635	11576	25943	1	1	0.0281617	0.0237058	3.3884
7	246.551	9801	24158	1	1	0.0285685	0.0257058	3.2448
8	224.693	8620	22726	1	0.995	0.0134571	0.00752442	0.8901
9	188.93	7444	20339	1	1	0.0386889	0.0284675	4.2344
10	120.125	5980	17550	1	1	0.0362876	0.0298576	6.6597
11	144.368	6541	18659	1	1	0.0067268	0.00482341	0.6927
12	204.995	8412	22105	1	0.9866	0.0224679	0.0186836	2.9269

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	965	1645	1114.49	2032.91	24	20	24
2	157	270	249.297	545.362	3	6	7
3	180	306	212.51	490.09	4	4	6
4	469	771	544.174	1084.18	14	12	12
5	377	561	457.789	826.281	10	9	8
6	258	483	381.747	709.24	5	9	9
7	264	483	322.735	712.826	10	8	10
8	30	30	123.74	184.853	1	3	2
9	233	399	333.806	676.926	8	9	7
10	142	266	247.894	589.659	3	7	8
11	40	49	47.799	95.0122	1	1	2
12	164	266	211.752	459.919	4	5	4


536-63x_Hyp_RD_z03c1-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 21296)
count
width (px)